

Notes: Carhart (JF, 1997)
On Persistence in Mutual Fund Performance

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1 Big picture

1.1 Question

Carhart asks whether persistence in equity mutual fund performance reflects genuine managerial skill or whether it can be explained by common stock-return factors and fund costs. The paper revisits the empirical fact that recent winner funds tend to keep doing well for a short time and recent loser funds tend to keep doing poorly, then asks what economic mechanism generates this pattern.

The central answer is that persistence in mutual fund returns is mostly not stock-picking skill. Much of the winner-loser spread comes from exposure to common return factors, especially the one-year momentum factor. Much of the remaining spread comes from expenses, turnover-related transaction costs, and load-related costs. The unexplained part is concentrated mainly in severe underperformance among the worst funds.

1.2 Setting

The paper studies diversified U.S. equity mutual funds from January 1962 through December 1993. A key strength is that the database includes nonsurviving funds, so the results are not mechanically biased upward by dropping failed or merged funds. Returns are reported net of fund operating expenses and security-level trading costs, but not net of sales loads.

Carhart sorts mutual funds into portfolios by past performance and then evaluates subsequent performance using factor models. The most important benchmark is the four-factor model that adds a one-year momentum factor to the Fama-French market, size, and value factors.

1.3 Core finding

A strategy that buys last year's top-decile mutual funds and sells last year's bottom-decile funds earns a large average return spread, roughly 8 percent per year. But the four-factor model explains a large part of this spread. Momentum exposure is central: last year's winning funds tend to hold more of last year's winning stocks, and last year's losing funds tend to hold more of last year's losing stocks.

Once common factors, expense ratios, turnover costs, and load-related costs are accounted for, the remaining evidence for skill is weak. The main anomaly is not repeated superior performance by winners, but persistent poor performance by the worst-performing funds.

1.4 Economic takeaway

The paper shifts the interpretation of mutual fund persistence from "manager skill" to "strategy plus costs." Winners often look good because their portfolios load on momentum at the right time. Losers often keep underperforming because they are costly, turn over more, and may be poorly managed or inefficient. Investors should be cautious about chasing funds with one good year of returns, but should pay close attention to persistent poor performance and high costs.

2 Incremental contribution of the paper

2.1 Relative to prior mutual fund persistence papers

Earlier work documented short-horizon persistence and described it as a “hot hands” effect. Carhart shows that this hot-hands pattern is largely explained by the one-year stock momentum effect documented by Jegadeesh and Titman. The result is important because it separates fund-level persistence from manager-level skill.

2.2 Relative to traditional performance measurement

The paper introduces what became the standard Carhart four-factor model for mutual fund performance:

$$r_{it} = \alpha_i + b_i RMRF_t + s_i SMB_t + h_i HML_t + p_i PR1YR_t + e_{it}.$$

The key addition is *PR1YR*, a factor-mimicking portfolio for one-year momentum. This factor is essential because mutual funds can mechanically display return persistence if their holdings are tilted toward recent winners or losers.

2.3 Relative to cost-based explanations

Carhart makes fund costs central. Expense ratios, turnover, and load fees are not just background variables; they directly reduce net performance and explain part of the observed persistence. This is especially important because costs are themselves persistent across funds.

2.4 Relative to investor choice

The paper gives a practical interpretation of persistence. Chasing last year’s winners can capture some momentum exposure, but this advantage is short-lived. Avoiding high-cost and persistently poor funds is more robust. The paper therefore links performance measurement to investor portfolio choice.

3 Data and measurement

3.1 Survivor-bias-free mutual fund database

The database covers 1,892 diversified equity funds and 16,109 fund-years. It includes funds that disappear before the end of the sample, which matters because unsuccessful funds are more likely to close, merge, or stop being observable. Excluding dead funds would overstate average performance and distort persistence tests.

The average fund has total net assets of about \$218 million, an expense ratio of 1.14 percent per year, and modified turnover of 77.3 percent per year. Dead funds are smaller, have lower flows, higher expenses, and higher turnover than live funds. This already suggests that costs and survivorship are important to any study of fund performance.

3.2 Return definitions

Let R_{it} denote the return on fund or portfolio i , and let R_{ft} be the one-month Treasury bill return. The excess return is

$$r_{it} = R_{it} - R_{ft}.$$

The paper evaluates whether r_{it} is unusually high after controlling for systematic return factors. The intercept α_i is the abnormal return or performance estimate.

3.3 Factor models

The CAPM benchmark is

$$r_{it} = \alpha_i + \beta_i VWRF_t + \varepsilon_{it},$$

where $VWRF_t$ is the excess return on the CRSP value-weighted market portfolio.

The Fama-French three-factor model is

$$r_{it} = \alpha_i + b_i RMRF_t + s_i SMB_t + h_i HML_t + \varepsilon_{it},$$

where $RMRF_t$ is the market excess return, SMB_t is the size factor, and HML_t is the value factor.

The Carhart four-factor model adds momentum:

$$r_{it} = \alpha_i + b_i RMRF_t + s_i SMB_t + h_i HML_t + p_i PR1YR_t + \varepsilon_{it}.$$

Here $PR1YR_t$ is the return on a portfolio long recent winners and short recent losers, using prior eleven-month returns lagged one month.

3.4 Why the momentum factor matters

Suppose a fund has positive loading p_i on $PR1YR_t$. If momentum earns a positive average premium, then the fund will have high expected returns even without stock-picking skill. Omitting $PR1YR_t$ would mistake momentum exposure for manager ability.

For a portfolio spread between past winners and past losers, expected excess returns satisfy

$$E[r_{1-10}] = \alpha_{1-10} + \Delta b E[RMRF] + \Delta s E[SMB] + \Delta h E[HML] + \Delta p E[PR1YR].$$

The term $\Delta p E[PR1YR]$ is large in Carhart's results. This is the mechanical reason why the four-factor alpha spread is much smaller than the raw return spread.

4 Empirical design

4.1 Short-horizon persistence portfolios

At the beginning of each year, mutual funds are sorted into deciles based on their prior calendar-year returns. The portfolios are equal-weighted and held for one year. Funds that disappear during the year remain in the portfolio until disappearance, and then weights are adjusted. This procedure preserves the impact of poor funds that exit the sample.

Funds in decile 1 are last year's winners; funds in decile 10 are last year's losers. The paper also subdivides the top and bottom deciles into thirds, making portfolios 1A, 1B, 1C and 10A, 10B, 10C.

4.2 Cross-sectional cost regressions

To evaluate the impact of fund characteristics on performance, Carhart estimates monthly cross-sectional Fama-MacBeth regressions of residual performance on fund characteristics:

$$\alpha_{it} = a_t + b_t x_{it} + \varepsilon_{it}.$$

The dependent variable is the monthly residual from the four-factor model. The key explanatory variables include expense ratio, turnover, log total net assets, maximum load, buy turnover, and sell turnover.

The interpretation is direct: if the coefficient on expenses is negative, then higher reported expense ratios reduce net abnormal performance. If the coefficient on turnover is negative, then trading activity consumes returns through transaction costs.

4.3 Longer-horizon alpha sorts

The paper also sorts funds by past three-year four-factor alpha estimates. This test asks whether past risk-adjusted performance predicts future risk-adjusted performance better than raw returns do. It is a stricter test of managerial skill, but it also has a joint-hypothesis problem: the same model is used both to rank funds and to evaluate subsequent performance.

5 Main results

5.1 One-year persistence is mostly factor exposure

The top past-return funds outperform the bottom past-return funds by about 0.67 percent per month. Under the CAPM, the alpha spread is nearly the same, because the winner and loser portfolios have similar market betas. The CAPM therefore leaves the appearance of strong abnormal performance.

The four-factor model changes the interpretation. The 1-minus-10 alpha spread falls to about 0.29 percent per month. Much of the winner-loser spread comes from the momentum factor. The top portfolio loads positively on *PR1YR*, while the bottom portfolio loads negatively. This means that recent mutual fund winners tend to be exposed to recent stock winners.

5.2 The worst funds are special

The largest unexplained pattern is concentrated at the bottom. The spread between deciles 9 and 10 is a large fraction of the remaining unexplained performance. The bottom funds also have high expense ratios and high turnover. This supports the paper's conclusion that persistent underperformance is more robust than persistent superior performance.

5.3 Expenses and turnover reduce performance

The Fama-MacBeth regressions show that expense ratios have a strongly negative coefficient. The estimate is about -1.54 with a t -statistic of -5.99 , implying that expenses reduce performance by at least one-for-one.

Turnover is also negatively related to performance. The modified turnover coefficient is about -0.95 , which the paper interprets as an approximate round-trip trading cost of 95 basis points per traded dollar. Sell turnover is especially costly, with a coefficient of -1.26 and a t -statistic of -3.00 .

5.4 Load funds underperform

Maximum load fees are negatively related to subsequent performance, even after accounting for reported expense ratios. Carhart interprets this as evidence that load funds do not compensate investors with superior management. Instead, load funds appear to underperform no-load funds by roughly 80 basis points per year, holding other costs fixed.

5.5 Three-year alpha persistence is weak evidence of skill

Sorting funds by three-year past four-factor alpha produces a positive alpha spread between past winners and losers. But the best alpha funds do not earn reliably positive future alphas. Their performance is only relatively better than other funds. This is consistent with some top funds covering their costs, but not with strong evidence of persistent stock-picking skill.

6 Mechanism and interpretation

6.1 Momentum versus skill

A fund can look persistent because it loads on a persistent stock-return factor. If last year's winning stocks keep winning for a while, then a fund with a portfolio tilted toward those stocks will also perform well. This is not the same as evidence that the manager has superior private information.

The four-factor model is designed to avoid this mistake. It attributes to the manager only the part of performance not explained by market, size, value, and momentum factor exposures.

6.2 Costs as persistent performance drag

Costs are persistent and differ across funds. High-expense, high-turnover funds tend to remain high-expense, high-turnover funds. Therefore, cost differences naturally generate persistence in net returns. This mechanism is especially important because mutual fund returns are reported net of expenses and trading costs.

6.3 Why losers persist more than winners

Superior performance is hard to repeat once momentum exposure fades. Poor performance is more persistent because it can be generated by ongoing cost drag, high turnover, poor execution, and weak portfolio choices. This asymmetry explains why the bottom tail is more important than the top tail in Carhart's evidence.

7 Detailed derivations

7.1 Four-factor alpha and expected return

The four-factor regression is

$$r_{it} = \alpha_i + b_i RMRF_t + s_i SMB_t + h_i HML_t + p_i PR1YR_t + e_{it}.$$

Taking expectations gives

$$E[r_i] = \alpha_i + b_i E[RMRF] + s_i E[SMB] + h_i E[HML] + p_i E[PR1YR].$$

Thus, the alpha is

$$\alpha_i = E[r_i] - b_i E[RMRF] - s_i E[SMB] - h_i E[HML] - p_i E[PR1YR].$$

A high raw return does not imply a high alpha if it comes from high loadings on rewarded factors.

7.2 Winner-loser spread decomposition

Let portfolio W be the past winner portfolio and L be the past loser portfolio. The expected spread is

$$E[r_W - r_L] = (\alpha_W - \alpha_L) + (b_W - b_L)E[RMRF] + (s_W - s_L)E[SMB] + (h_W - h_L)E[HML] + (p_W - p_L)E[PR1YR].$$

In Table III, the raw 1-minus-10 spread is 0.67 percent per month, while the four-factor alpha spread is 0.29 percent per month. Hence approximately

$$0.67\% - 0.29\% = 0.38\%$$

per month is explained by factor exposure. Since the $PR1YR$ loading spread is about 0.38 and the average monthly $PR1YR$ return is about 0.82 percent, momentum alone explains roughly

$$0.38 \times 0.82\% \approx 0.31\%$$

per month of the winner-loser spread.

7.3 Expense ratio effect

If the Fama-MacBeth regression coefficient on expense ratio is -1.54 , then a higher expense ratio lowers net performance more than one-for-one. Conceptually, if a fund increases its expense ratio by 1 percentage point per year, the expected reduction in annual abnormal performance is approximately 1.54 percentage points, before considering indirect effects through turnover or loads.

7.4 Turnover and transaction costs

The coefficient on modified turnover is -0.95 . Since modified turnover measures the trading intensity of the fund, the coefficient implies that a round-trip trade costs nearly 0.95 percent of the traded market value. This converts turnover into a persistent drag on performance:

$$\text{trading drag} \approx 0.0095 \times \text{modified turnover}.$$

For funds with high annual turnover, this effect is economically meaningful.

7.5 Adjusted alpha in the long-horizon tests

Table VI reports an adjusted alpha that adds back expense ratios and estimated transaction costs:

$$\alpha^{adj} = \alpha + \frac{1}{12} \text{Expense Ratio} + \frac{1}{12} (\text{Turnover} \times \text{Roundtrip Transaction Cost}).$$

This calculation asks whether poor performance is merely the mechanical effect of costs. Even after this adjustment, the bottom funds remain noticeably weak, which is why Carhart emphasizes persistent underperformance.

8 Extracted figures and tables from the paper

8.1 Table I: Mutual Fund Database Summary Statistics; Table II: Performance Measurement Model Summary Statistics, July 1963 to December 1993

Table I
Mutual Fund Database Summary Statistics

The table reports time-series averages of annual cross-sectional averages from 1962 to 1993. TNA is total net assets, Flow is the percentage change in TNA adjusted for investment return and mutual fund mergers. Exp ratio is total annual management and administrative expenses divided by average TNA. Mturn is modified turnover and represents reported turnover plus 0.5 times Flow. Maximum load is the total of maximum front-end, rear-end, and deferred sales charges as a percentage of the investment. Live funds are those in operation at the end of the sample, December 31, 1993. Dead funds are those that discontinued operations prior to this date.

Time-Series Averages of Cross-Sectional Average Annual Attributes, 1962–1993									
Group	Total Number	Avg Number	Avg TNA (\$ millions)	Avg Flow (%/year)	Avg Exp Ratio (%/year)	Avg Mturn (%/year)	Percentage with Load	Avg Max Load	Avg Age (years)
All funds	1,892	509.1	\$217.8	3.4%	1.14%	77.3%	64.5%	7.33%	18.1
By fund category									
Aggressive growth	675	169.2	\$ 95.6	5.0%	1.55%	99.7%	58.2%	7.38%	12.3
Long term growth	618	168.5	\$221.4	5.5%	1.09%	79.5%	59.7%	7.38%	16.4
Growth & income	599	171.4	\$328.5	1.5%	0.91%	60.9%	70.0%	7.27%	23.8
By current status									
Live funds	1,310	352.3	\$268.7	4.3%	1.07%	76.2%	63.3%	7.29%	19.2
Dead funds	582	156.8	\$ 46.8	–1.2%	1.44%	83.1%	68.5%	7.44%	14.9

Table I: Mutual Fund Database Summary Statistics

Table II
Performance Measurement Model Summary Statistics, July 1963 to December 1993

VWRF is the Center for Research in Security Prices (CRSP) value-weight stock index minus the one-month T-bill return. RMRF is the excess return on Fama and French's (1993) market proxy. SMB and HML are Fama and French's factor-mimicking portfolios for size and book-to-market equity. PRIYR is a factor-mimicking portfolio for one-year return momentum.

Factor Portfolio	Monthly			Cross-Correlations				
	Excess Return	Std Dev	t-stat for Mean = 0	VWRF	RMRF	SMB	HML	PRIYR
VWRF	0.44	4.39	1.93	1.00				
RMRF	0.47	4.43	2.01	1.00	1.00			
SMB	0.29	2.89	1.89	0.35	0.32	1.00		
HML	0.46	2.59	3.42	–0.36	–0.37	0.10	1.00	
PRIYR	0.82	3.49	4.46	0.01	0.01	–0.29	–0.16	1.00

Table II: Performance Measurement Model Summary Statistics, July 1963 to December 1993

Interpretation of Table I. The survivor-bias-free database is central. It includes both live and dead funds. Dead funds are smaller, have negative flows, higher expenses, and higher turnover. This confirms why survivor-bias correction matters: funds that disappear are not a random subset of the population.

Interpretation of Table II. The momentum factor has a high average return, 0.82 percent per month, and low correlation with the market factors. This makes it a powerful additional explanatory factor rather than a redundant repackaging of market, size, or value exposure.

8.2 Table III: Portfolios of Mutual Funds Formed on Lagged 1-Year Return; Table IV: Characteristics of the Portfolios of Mutual Funds Formed on Lagged 1-Year Return

Table III

Portfolios of Mutual Funds Formed on Lagged 1-Year Return

Mutual funds are sorted on January 1 each year from 1963 to 1993 into decile portfolios based on their previous calendar year's return. The portfolios are equally weighted monthly so the weights are readjusted whenever a fund disappears. Funds with the highest past one-year return comprise decile 1 and funds with the lowest comprise decile 10. Deciles 1 and 10 are further subdivided into thirds on the same measure. VWRF is the excess return on the CRSP value-weight market proxy. RMRF, SMB, and HML are Fama and French's (1993) market proxy and factor-mimicking portfolios for size and book-to-market equity. PR1YR is a factor-mimicking portfolio for one-year return momentum. Alpha is the intercept of the Model. The t-statistics are in parentheses.

Portfolio	Monthly		CAPM			4-Factor Model					
	Excess Return	Std Dev	Alpha	VWRF	Adj R-sq	Alpha	RMRF	SMB	HML	PR1YR	Adj R-Sq
1A	0.75%	5.45%	0.27%	1.08	0.777	-0.11%	0.91	0.72	-0.07	0.33	0.891
			(2.06)	(35.94)		(-1.11)	(37.67)	(19.95)	(-1.65)	(11.53)	
1B	0.67%	4.94%	0.22%	1.00	0.809	-0.10%	0.86	0.59	-0.05	0.27	0.898
			(2.00)	(39.68)		(-1.08)	(40.66)	(18.47)	(-1.38)	(10.63)	
1C	0.63%	4.95%	0.17%	1.02	0.843	-0.15%	0.89	0.56	-0.05	0.27	0.927
			(1.70)	(44.65)		(-1.92)	(49.76)	(20.86)	(-1.61)	(12.69)	
1 (high)	0.68%	5.04%	0.22%	1.03	0.834	-0.12%	0.88	0.62	-0.05	0.29	0.933
			(2.10)	(43.11)		(-1.60)	(50.54)	(23.67)	(-1.86)	(13.88)	
2	0.59%	4.72%	0.14%	1.01	0.897	-0.10%	0.89	0.46	-0.05	0.20	0.955
			(1.75)	(57.00)		(-1.78)	(66.47)	(22.95)	(-2.25)	(12.43)	
3	0.43%	4.56%	-0.01%	0.99	0.931	-0.18%	0.90	0.34	-0.07	0.16	0.963
			(-0.08)	(70.96)		(-3.65)	(76.80)	(18.99)	(-3.69)	(11.52)	
4	0.45%	4.41%	0.02%	0.97	0.952	-0.12%	0.90	0.27	-0.05	0.11	0.971
			(0.33)	(85.70)		(-2.81)	(90.03)	(18.18)	(-3.12)	(9.40)	
5	0.38%	4.35%	-0.05%	0.96	0.960	-0.14%	0.90	0.22	-0.05	0.07	0.970
			(-1.10)	(93.93)		(-3.31)	(89.65)	(14.42)	(-3.27)	(6.18)	
6	0.40%	4.36%	-0.02%	0.96	0.958	-0.12%	0.90	0.22	-0.04	0.08	0.968
			(-0.46)	(91.94)		(-2.82)	(86.16)	(14.02)	(-2.37)	(6.01)	
7	0.36%	4.30%	-0.06%	0.95	0.959	-0.14%	0.90	0.21	-0.03	0.04	0.967
			(-1.39)	(92.90)		(-3.09)	(85.73)	(13.17)	(-1.62)	(2.89)	
8	0.34%	4.48%	-0.10%	0.98	0.951	-0.13%	0.93	0.20	-0.06	0.01	0.958
			(-1.86)	(85.14)		(-2.52)	(75.44)	(10.74)	(-3.16)	(0.84)	
9	0.23%	4.60%	-0.21%	1.00	0.926	-0.20%	0.93	0.22	-0.10	-0.02	0.938
			(-3.24)	(67.91)		(-3.11)	(60.44)	(9.69)	(-3.80)	(-1.17)	
10 (low)	0.01%	4.90%	-0.45%	1.02	0.851	-0.40%	0.93	0.32	-0.08	-0.09	0.887
			(-4.58)	(46.09)		(-4.33)	(42.23)	(9.69)	(-2.23)	(-3.50)	
10A	0.25%	4.78%	-0.19%	1.00	0.864	-0.19%	0.91	0.33	-0.11	-0.02	0.891
			(-2.05)	(48.48)		(-2.16)	(42.99)	(10.27)	(-3.20)	(-0.76)	
10B	0.02%	4.92%	-0.42%	1.00	0.817	-0.37%	0.91	0.32	-0.09	-0.09	0.848
			(-3.84)	(40.67)		(-3.45)	(35.52)	(8.24)	(-2.16)	(-2.99)	
10C	-0.25%	5.44%	-0.74%	1.05	0.736	-0.64%	0.98	0.32	-0.04	-0.17	0.782
			(-5.06)	(32.16)		(-4.49)	(28.82)	(6.29)	(-0.73)	(-4.09)	
1-10 spread	0.67%	2.71%	0.67%	0.01	-0.002	0.29%	-0.05	0.30	0.03	0.38	0.231
			(4.68)	(0.39)		(2.13)	(-1.52)	(6.30)	(0.53)	(10.07)	
1A-10C spread	1.01%	3.87%	1.00%	0.02	-0.002	0.53%	-0.07	0.40	-0.02	0.50	0.197
			(4.90)	(0.42)		(2.72)	(-1.61)	(5.73)	(0.32)	(8.98)	
9-10 spread	0.23%	1.22%	-0.02	0.004	0.20%	-0.01	-0.10	-0.01	0.07	0.118	
			(3.64)	(-1.60)		(3.13)	(-0.40)	(-4.30)	(-0.60)	(3.87)	

Table III: Portfolios of Mutual Funds Formed on Lagged 1-Year Return

Table IV

Characteristics of the Portfolios of Mutual Funds Formed on Lagged 1-Year Return

Mutual funds are sorted annually from 1963 to 1993 into equal-weight decile portfolios based on lagged one-year return. Funds with the highest past one-year return comprise decile 1, and funds with the lowest comprise decile 10. Deciles 1 and 10 are further subdivided into thirds on the same measure. The values in the table represent the time-series averages of annual cross-sectional averages of the funds in each portfolio. TNA is total net assets. Expense ratio is management, administrative, and 12b-1 expenses divided by average TNA. Mturn is modified turnover and represents reported turnover plus 0.5 times the percentage change in portfolio TNA adjusted for investment returns and mergers. Maximum load is the sum of maximum front-end, back-end, and deferred sales charges.

Average Annual Portfolio Attributes					
Portfolio	Age (years)	TNA (\$ millions)	Expense Ratio	Mturn	Maximum Load
1A	11.7	110.0	1.38	116.2	3.93
1B	14.0	148.8	1.16	86.9	3.99
1C	16.5	127.4	1.11	75.8	4.62
1 (high)	14.1	128.7	1.22	92.9	4.18
2	16.6	190.8	1.08	75.3	4.97
3	17.3	194.3	1.10	76.3	4.72
4	17.6	183.7	1.11	67.2	4.82
5	18.3	185.9	1.09	68.4	4.71
6	17.5	199.1	1.15	65.8	4.33
7	18.3	169.7	1.14	62.2	4.50
8	17.5	149.3	1.13	65.3	4.76
9	15.8	145.6	1.22	75.1	4.59
10 (low)	13.6	77.1	1.92	81.4	4.38
10A	14.5	91.9	1.55	76.8	4.55
10B	14.4	87.4	1.71	76.7	4.57
10C	11.9	52.0	2.51	88.8	4.02

Table IV: Characteristics of the Portfolios Formed on Lagged 1-Year Return

Interpretation of Table III. Raw returns show strong persistence: the past winner decile earns much more than the past loser decile. CAPM alphas preserve this pattern because market beta does not differ much across deciles. The four-factor model sharply reduces the alpha spread because winners have positive momentum exposure and losers have negative momentum exposure.

Interpretation of Table IV. The lowest-ranked funds have much higher expense ratios and higher turnover than most other funds. The bottom third of loser funds, portfolio 10C, has especially high expenses and small assets. This helps explain why the paper views persistent poor performance as partly cost-based.

8.3 Table V: Fama-MacBeth (1973) Cross-Sectional Regressions; Figure 1: Contingency Table of Initial and Subsequent One-Year Performance Rankings

Table V
Fama-MacBeth (1973) Cross-Sectional Regressions

Estimated univariate cross-sectional regressions for each month from July 1966 to December 1993 across all funds in the sample at that time. The dependent variable is the monthly residual from the 4-factor model, where the factor loadings are estimated on the prior 3 years of monthly returns. The independent variables are expense ratio, turnover, the natural log of TNA, maximum load fees, and measures of buy and sell turnover. Expense ratio is management, administrative, and 12b-1 expenses divided by average TNA. TNA is total net assets. Turnover represents reported turnover plus 0.5 times the percentage change in portfolio TNA adjusted for investment returns and mergers. Maximum load is the sum of maximum front-end, back-end and deferred sales charges. Buy turnover is reported turnover plus the maximum of 0 and the percentage change in TNA adjusted for investment returns and mergers. Sell turnover is reported turnover minus the minimum of 0 and the adjusted percentage change in TNA. Expense ratio and the turnover measures are divided by 12 and measured contemporaneous with the dependent variable. The reported estimates are time-series averages of monthly cross-sectional regression slope estimates as in Fama and MacBeth (1973). The t -statistics are on the time-series means of the coefficients. The regressions on TNA, maximum load, and the turnover measures use the residuals from reported returns after adding back expense ratios.

Independent Variables (Coefficients $\times 100$)	Estimate	t -statistic
Expense ratio (t)	-1.54	(-5.99)
Turnover (t) (Mturn)	-0.95	(-2.36)
ln TNA (t-1)	-0.05	(-0.66)
Maximum Load (t-1)	-0.11	(-3.55)
Buy turnover (t)	-0.43	(-1.16)
Sell turnover (t)	-1.26	(-3.00)

Table V: Fama-MacBeth (1973) Cross-Sectional Regressions

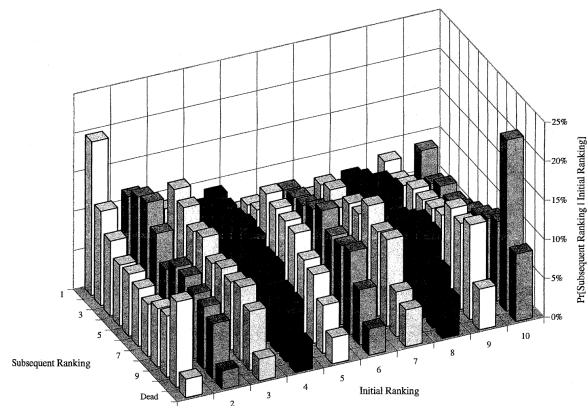


Figure 1. Contingency table of initial and subsequent one-year performance rankings. In each calendar year from 1962 to 1992, funds are ranked into decile portfolios based on one-year gross return. These initial decile rankings are paired with the fund's subsequent one-year gross return ranking. Funds that do not survive the complete subsequent year are placed in a separate category for dead funds. The bars in cell (j, i) represent the conditional probability of achieving a subsequent ranking of decile i (or divine) given an initial ranking of decile i . I estimate gross

Figure 1: Contingency Table of Initial and Subsequent One-Year Performance Rankings

Interpretation of Table V. Expense ratios and turnover have large negative coefficients. Expenses reduce performance more than one-for-one, and sell turnover is especially costly. The negative coefficient on maximum load also shows that load funds do not compensate investors through higher net performance.

Interpretation of Figure 1. Rank persistence is asymmetric. Winner funds are somewhat more likely to remain winners, but many revert to lower rankings. Loser funds are more likely to remain losers or disappear. The figure therefore supports a short-lived winner effect and a more persistent loser effect.

8.4 Figure 2: Post-Formation Returns on Portfolios of Mutual Funds Sorted on Lagged One-Year Return; Figure 3: Summary of Explanations for Persistence in Mutual Fund Performance

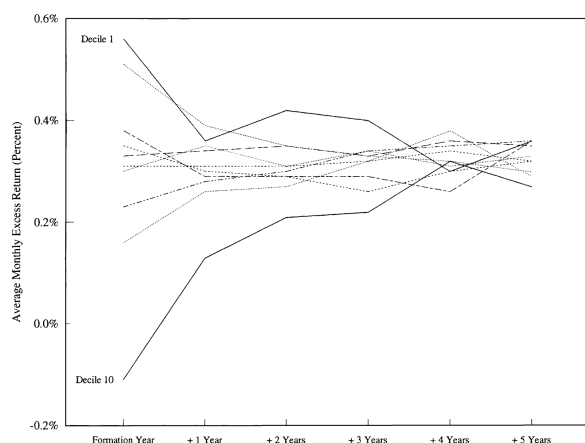


Figure 2. Post-formation returns on portfolios of mutual funds sorted on lagged one-year return. In each calendar year from 1962 to 1987, funds are ranked into equal-weight decile portfolios based on one-year return. The lines in the graph represent the excess returns on the

Figure 2: Post-Formation Returns on Lagged One-Year Return Portfolios

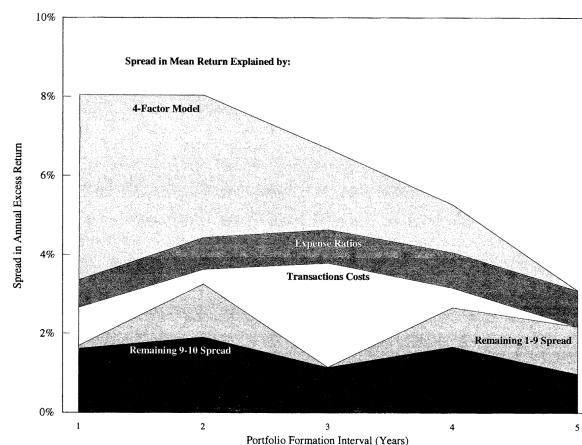


Figure 3. Summary of explanations for persistence in mutual fund performance. On January 1 of each year, funds are ranked into equal-weight decile portfolios based on returns over the prior one-, two-, three-, four-, and five-year periods. Funds with the highest return comprise decile 1, and funds with the lowest comprise decile 10. The height of the graph represents the annual spread in mean return between deciles 1 and 10 for the portfolios formed on one- to five-year returns. The top shaded region represents the spread in annual return that is explained by the 4-factor model, where the 4-factor model captures common variation in return associated with size, book-to-market equity, and one-year return momentum. The second region from the top represents the difference in the average expense ratios of deciles 1 and 10. The third region from

Figure 3: Summary of Explanations for Persistence in Mutual Fund Performance

Interpretation of Figure 2. The winner-loser return spread is primarily a one-year phenomenon. After the first post-ranking year, most decile returns converge. The main exception is the continued weakness of the worst funds.

Interpretation of Figure 3. Common factors explain the largest component of the short-horizon winner-loser spread. Expenses and transaction costs explain additional portions. The remaining unexplained spread is concentrated mostly between the ninth and tenth deciles rather than between the top and the middle of the distribution.

8.5 Table VI: Portfolios of Mutual Funds Formed on 3-Year Past 4-Factor Model Alphas; Figure 4: Post-Formation Returns on Portfolios of Mutual Funds Sorted on Lagged Three-Year Estimates of 4-Factor Alpha

Table VI
Portfolios of Mutual Funds Formed on 3-Year Past 4-Factor Model Alphas

Mutual funds are sorted on January 1 each year from 1966 to 1993 into equal-weight decile portfolios based on their 4-factor model alpha estimate over the prior 3 years. 1 requires a minimum of 30 return observations for this estimate. Funds with the highest alpha estimates comprise decile 1 and funds with the lowest comprise decile 10. The 4-factor model consists of the RMRP, SMB, and HML and PR1YR factor-mimicking portfolios. RMRP, SMB, and HML are Fama and French's (1993) market proxy and factor-mimicking portfolios for size and book-to-market equity. PR1YR is a factor-mimicking portfolio for one-year return momentum. Expense ratio and turnover are time-series averages of annual cross-sectional averages of the funds in each portfolio. Expense ratio is management, administrative, and 12b-1 expenses divided by average total net assets (TNA). Turnover represents reported turnover plus 0.5 times the percentage change in portfolio TNA adjusted for investment returns and mergers. Roundtrip transactions costs are estimated monthly, using cross-sectional regressions on turnover across all funds in each of 5 quintile sorts on lagged 4-factor alpha. The dependent variable in these regressions is the monthly residual from the 4-factor model, where the factor loadings are estimated on the prior 3 years of gross monthly returns after adding back expense ratios. Alpha is the 4-factor model intercept estimate, and alpha-*t* is the *t*-statistic on this estimate. Adjusted alpha is the 4-factor alpha plus 1/12 of expense ratio and 1/12 of turnover times roundtrip transaction costs.

Portfolio	Excess Return	Standard Deviation	4-Factor Model Ordinary Least Squares (OLS) Estimates								Exp Ratio	Turn (Mturn)	Roundtrip Transaction Costs	Adjusted Alpha
			Alpha	Alpha- <i>t</i>	RMRP	SMB	HML	PR1YR						
1 (high)	0.62%	5.07%	0.02%	(0.41)	0.93	0.48	-0.14	0.14	1.13	91.1	-0.17%	0.10%		
2	0.47%	4.60%	-0.06%	(-1.37)	0.90	0.32	-0.10	0.10	1.00	69.5	-0.17%	0.01%		
3	0.49%	4.49%	-0.03%	(-0.81)	0.90	0.25	-0.06	0.09	0.94	63.7	0.96%	0.19%		
4	0.43%	4.43%	-0.05%	(-1.46)	0.91	0.20	-0.04	0.06	0.95	58.6	0.96%	0.08%		
5	0.39%	4.45%	-0.13%	(-3.34)	0.90	0.25	-0.03	0.09	0.97	57.4	0.94%	-0.01%		
6	0.40%	4.40%	-0.11%	(-2.76)	0.90	0.20	-0.03	0.08	0.98	61.1	0.84%	0.01%		
7	0.38%	4.46%	-0.17%	(-3.97)	0.90	0.26	-0.01	0.10	1.10	68.7	1.03%	-0.02%		
8	0.40%	4.54%	-0.16%	(-3.29)	0.90	0.29	-0.03	0.11	1.11	64.4	1.03%	-0.01%		
9	0.37%	4.68%	-0.19%	(-3.19)	0.89	0.38	-0.06	0.10	1.26	72.4	1.24%	-0.01%		
10 (low)	0.19%	5.10%	-0.43%	(-5.89)	0.93	0.49	-0.06	0.11	1.76	96.1	1.24%	-0.18%		
1-10 spread	0.43%	1.33%	0.45%	(3.95)	0.00	-0.01	-0.08	0.03	-0.63	-5.0	-1.41%	0.29%		
9-10 spread	0.18%	1.07%	0.24%	(3.12)	-0.04	-0.11	-0.01	-0.50	-23.7	NA	0.17%	NA		

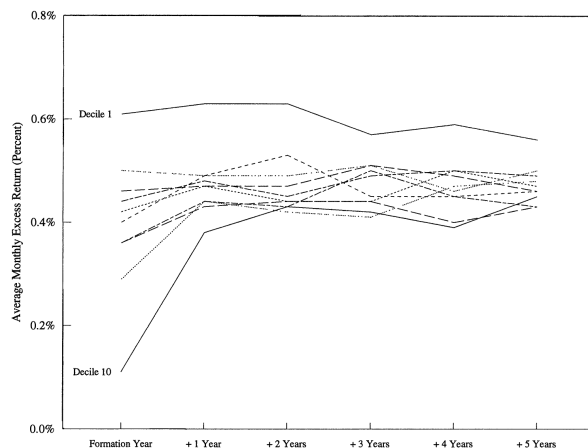


Figure 4. Post-formation returns on portfolios of mutual funds sorted on lagged three-year estimates of 4-factor alpha. In each calendar year from 1962 to 1987, funds are ranked into equal-weight decile portfolios based on three-year estimates of 4-factor alpha. The lines in the graph represent the excess returns on the decile portfolios in the year subsequent to initial ranking (the "formation" year) and each of the next five years after formation. Funds with the highest 4-factor alpha comprise decile 1, and funds with the lowest comprise decile 10. The portfolios are equally weighted each month, so the weights are readjusted whenever a fund disappears from the

Table VI: Portfolios Formed on 3-Year Past 4-Factor Model Alphas

Figure 4: Post-Formation Returns on Lagged Three-Year 4-Factor Alpha Portfolios

Interpretation of Table VI. Sorting on past alpha produces a larger future alpha spread than sorting on raw returns, but the top past-alpha funds still do not earn strongly positive future alphas. The lowest past-alpha funds continue to underperform strongly, even after adding back expenses and estimated transaction costs.

Interpretation of Figure 4. The top alpha-ranked funds stay above the middle portfolios for several years, but the more striking visual pattern is the recovery yet persistent weakness of the bottom alpha-ranked funds. The figure reinforces the conclusion that performance persistence is stronger in the left tail than in the right tail.

9 How the results fit together

9.1 Observed persistence

The starting empirical fact is that past winners outperform past losers. This fact is real in raw returns, and it is economically large.

9.2 Factor explanation

The first layer of explanation is common factor exposure. Winner funds load more on momentum and small stocks. Once *PR1YR* is included, the alpha spread shrinks considerably.

9.3 Cost explanation

The second layer is cost. Expense ratios, turnover, and loads reduce realized fund returns. Because these costs are persistent, they create persistence in net performance.

9.4 Residual pattern

The remaining unexplained persistence is not symmetric. It is concentrated among the worst funds. This matters for interpretation: the evidence is stronger for identifying bad funds than for identifying skilled funds.

10 Limitations and caveats

10.1 The joint-hypothesis problem

Any alpha estimate is conditional on the chosen asset-pricing model. A fund that appears to have positive alpha under one model may simply have exposure to a missing factor. Carhart's four-factor model improves on the CAPM and three-factor model, but the interpretation of alpha remains model-dependent.

10.2 Momentum is modeled as a factor, not explained structurally

The paper uses momentum as a benchmark factor. It does not fully explain why the momentum premium exists. Therefore, the paper can show that fund persistence is explained by momentum exposure, but it does not settle whether momentum itself is risk-based or mispricing-based.

10.3 Holdings data are not the focus

The analysis primarily uses fund returns and characteristics rather than detailed holdings. The conclusion about momentum exposure comes from return loadings, not from a full reconstruction of every fund's portfolio trades.

10.4 Investor implementation costs

Buying past winner funds may be cheaper than trading the underlying momentum portfolio directly, but the strategy could create flow-driven trading costs for fund shareholders. Carhart notes that mutual funds may need transaction fees for incoming and outgoing investors when such strategies become widely used.

11 Conclusion

Carhart's paper provides a cost-and-factor explanation for mutual fund performance persistence. The key empirical object is not raw persistence alone, but persistence after controlling for market, size, value, momentum, expenses, turnover, and loads.

The main conclusion is that mutual fund performance persistence does not generally reflect superior stock-picking skill. The apparent hot-hands result is largely a momentum effect. Expenses and transaction costs further explain why some funds persistently underperform. The most robust investor lesson is to avoid funds with persistently poor performance and high costs.